

Youwei Xiao

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Education

Bachelor of Science degree in Computer Science

School of Electronics Engineering and Computer Science, Peking University

September 2018 - July 2022

Beijing, China

Ph.D. Candidate in EDA

School of Integrated Circuits, Peking University

September 2022 - now

Beijing, China

- Research: compiler-driven hardware-software co-design for agile chips.

Work Experience

Large Language Model Algorithm Intern

ByteDance Seed

June 2026 – Present

Beijing, China

- Created an extensible benchmark of kernel programming and optimization tasks for dataflow-architecture chips.
- Developed a layered knowledge and skill system across multiple compilation paths and full-spectrum data collection and training workflows for continual model evolution.
- Pursuing the ultimate goal of developing a methodology and workflow that enable models and agents to rapidly acquire expert-level kernel optimization capabilities for new chip architectures.

Research

My research follows a compiler-driven approach to agile hardware-software co-design. Three coupled threads form a closed loop: architecture description and exploration discovers and formalizes reusable hardware capabilities; chip DSL and compilation express those capabilities and lower them into implementations; compiler optimization and LLM systems improve the resulting compiler, execution, and co-design workflows. My current work creates an extensible benchmark of kernel programming and optimization tasks for dataflow-architecture chips, develops a layered knowledge and skill system across multiple compilation paths, and builds full-spectrum data collection and training workflows for continual model evolution. The ultimate goal is a methodology and workflow that enable models and agents to rapidly acquire expert-level kernel optimization capabilities for new chip architectures.

Internship topic: Agent for automated chip-kernel optimization across new domains and long-horizon tasks

Extensible dataflow-chip kernel benchmark | Multi-path layered knowledge & skills | Full-spectrum data & training workflows

Research work

Architecture description & exploration

Analysis, formal methods, and DSLs describe and explore architecture design spaces.

ISAMORE (ASPLOS '26)

Cayman (DAC '25)

APS (ICCAD '25)

Aquas (ICCAD '26)

Chip DSL & compilation

DSLs, multi-level IRs, and compiler stacks lower architecture intent into hardware.

Hector (ICCAD '22)

Cement (FPGA '24)

Cement2 (preprint)

Clay (ICCAD '25)

SkyEgg (preprint)

Compiler optimization & LLM systems

LLMs advance compiler/co-design workflows; compiler/runtime systems support LLM execution.

EggMind (preprint)

IntelliC (GitHub)

Spine (GitHub)

PTO Runtime (GitHub)

Hive (preprint)

Architecture description and exploration. I combine compiler analysis, formal methods, and architecture DSLs to represent and search design spaces while connecting customized hardware capabilities to programmable software. *ISAMORE* (DOI) discovers reusable custom instructions by anti-unifying equivalent program fragments; *Cayman* (DOI)

identifies accelerator opportunities in complete applications while co-optimizing control flow and data access; *APS* (DOI, GitHub) and *Aquas* (ICCAD 2026) describe customized architectures through DSLs and expose them through complete hardware/software compiler stacks.

Chip DSL and compilation. I create DSLs and intermediate representations at multiple abstraction levels, then develop compilation and synthesis passes that optimize timing, microarchitecture, implementation selection, mapping, and scheduling. *Hector* (DOI, GitHub) provides multi-level MLIR foundations; *Cement* (DOI) and *Cement2* (preprint) introduce cycle-deterministic DSL and temporal-transaction abstractions; *Clay* (DOI) makes ASIP instructions microarchitecture-aware; *SkyEgg* (preprint) jointly optimizes implementation selection and scheduling with e-graphs.

Compiler optimization and LLM systems. I explore LLMs for compiler and hardware-software co-design workflows, together with compiler and system support for LLM workloads. *EggMind* (preprint) synthesizes equality-saturation strategies; *IntelliC* (GitHub) studies inspectable representations for human-model compilation; *Spine* (GitHub) organizes verification-bounded cross-layer co-synthesis; *PTO Runtime* (GitHub) supports dynamic kernel fusion and distributed execution of compiled task graphs on Ascend chips and LingQu SuperPods; *Hive* (preprint) provides multi-agent inference infrastructure.

Publications

- **Youwei Xiao**, Chenyun Yin, Yitian Sun, and Yun Liang. 2026. Finding Reusable Instructions via E-Graph Anti-Unification. In Proceedings of the 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS '26). doi:10.1145/3779212.3790162. **[Best Paper Award, 5/1048]**
- Yuyang Zou, **Youwei Xiao**, Yansong Xu, Chenyun Yin, Yuhao Luo, Yitian Sun, Ruifan Xu, Renze Chen, and Yun Liang. 2026. Aquas: Enhancing Domain Specialization through Holistic Hardware-Software Co-Optimization based on MLIR. In Proceedings of the 45th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '26).
- **Youwei Xiao**, Yuyang Zou, Yansong Xu, Yuhao Luo, Yitian Sun, Chenyun Yin, Ruifan Xu, Renze Chen, and Yun Liang. 2025. APS: Open-Source Hardware-Software Co-Design Framework for Agile Processor Specialization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240817. [Invited Paper]
- Weijie Peng*, **Youwei Xiao***, Yuyang Zou, Zizhang Luo, and Yun Liang. 2025. Clay: High-level ASIP Framework for Flexible Microarchitecture-Aware Instruction Customization. In Proceedings of the 44th IEEE/ACM International Conference on Computer-Aided Design (ICCAD '25). doi:10.1109/ICCAD66269.2025.11240669. (*Equal contribution)
- **Youwei Xiao**, Fan Cui, Zizhang Luo, Weijie Peng, and Yun Liang. 2025. Cayman: Custom Accelerator Generation with Control Flow and Data Access Optimization. In Proceedings of the 62nd ACM/IEEE Design Automation Conference (DAC '25). doi:10.1109/DAC63849.2025.11132875.
- **Youwei Xiao**, Zizhang Luo, and Yun Liang. 2025. cmt2: Rule-Based Hardware Description in Rust with Temporal Semantics. In 5th Workshop on Languages, Tools, and Techniques for Accelerator Design (LATTE '25).
- Fan Cui, **Youwei Xiao**, Kexing Zhou, and Yun Liang. 2025. An Empirical Comparison of LLM-based Hardware Design and High-level Synthesis. In Proceedings of the 2025 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '25). doi:10.1145/3706628.3708861.
- Fan Cui, Chenyang Yin, Kexing Zhou, **Youwei Xiao**, Guangyu Sun, Qiang Xu, Qipeng Guo, Yun Liang, Xingcheng Zhang, Demin Song, and Dahua Lin. 2024. OriGen: Enhancing RTL Code Generation with Code-to-Code Augmentation and Self-Reflection. In Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design (ICCAD '24). doi:10.1145/3676536.3676830.
- **Youwei Xiao**, Zizhang Luo, Kexing Zhou, and Yun Liang. 2024. Cement: Streamlining FPGA Hardware Design with Cycle-Deterministic eHDL and Synthesis. In Proceedings of the 2024 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA '24). doi:10.1145/3626202.3637561.
- Ruifan Xu, **Youwei Xiao**, Jin Luo, and Yun Liang. 2022. HECTOR: A Multi-Level Intermediate Representation for Hardware Synthesis Methodologies. In Proceedings of the 41st IEEE/ACM International Conference on Computer-Aided Design (ICCAD '22). doi:10.1145/3508352.3549370.

Preprints

- Chenyun Yin*, **Youwei Xiao***, Yuze Luo, Yuyang Zou, and Yun Liang. LLM-Guided Strategy Synthesis for Scalable Equality Saturation. arXiv:2604.17364. (*Equal contribution)
- Zizhang Luo, Yuhao Luo, **Youwei Xiao**, Yansong Xu, Runlin Guo, and Yun Liang. Hive: A Multi-Agent Infrastructure for Algorithm- and Task-Level Scaling. arXiv:2604.17353.
- **Youwei Xiao**, Yuyang Zou, and Yun Liang. SkyEgg: Joint Implementation Selection and Scheduling for Hardware Synthesis using E-graphs. arXiv:2511.15323.
- **Youwei Xiao**, Zizhang Luo, Weijie Peng, Yuyang Zou, and Yun Liang. Cement2: Temporal Hardware Transactions for High-Level and Efficient FPGA Programming. arXiv:2511.15073.

Academic Service

- **Program committees:** Student Technical Program Committee Member, MICRO 2026; Technical Program Committee Member, MLCAD 2026.
- **Peer reviewer:** IEEE TCAD (2024); ACM TECS (2024).

Honors and Awards

学术之芯	2026
Awarded to 8 students for academic excellence	School of IC, Peking University
博士研究生校长奖学金	2026-2027
Awarded to 10 students in School of IC, Peking University	Peking University
ASPLOS 2026 Best Paper Award	March 2026
Finding Reusable Instructions via E-Graph Anti-Unification	5/1048 submissions
Second Place in EDATHON 2020	August 2020
Programming Competition on Electronic Design Automation	IEEE CEDA Hong Kong Chapter
Shenzhen Stock Exchange Scholarship	2019-2020
Merit Student	2018-2019 & 2019-2020
Awarded to top 10% students for comprehensive excellence	
Yang Fuqing & Wang Yangyuan Academician Scholarship	2018-2019
Awarded to only 5 students at EECS, Peking University, for academic excellence	
Second Prize in NOI 2017	July 2017
National Olympiad in Informatics	China Computer Federation
Second Prize in APIO 2017	May 2017
Asia Pacific Informatics Olympiad (China District)	China Computer Federation

Tutorials

APS: An MLIR-Based Hardware-Software Co-design Framework	January 2026
ASP-DAC 2026	Hong Kong, China
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design	May 2025
ISED 2025	Hong Kong, China
Agile Hardware Specialization (AHS)	March 2025
ASPLOS 2025	Rotterdam, Netherlands
Agile Hardware Specialization: A toolbox for Agile Chip Front-end Design	March 2025
DATE 2025	Lyon, France
AHS: An EDA toolbox for Agile Chip Front-end Design	January 2025
ASP-DAC 2025	Tokyo, Japan